1.11 Dirac Delta function

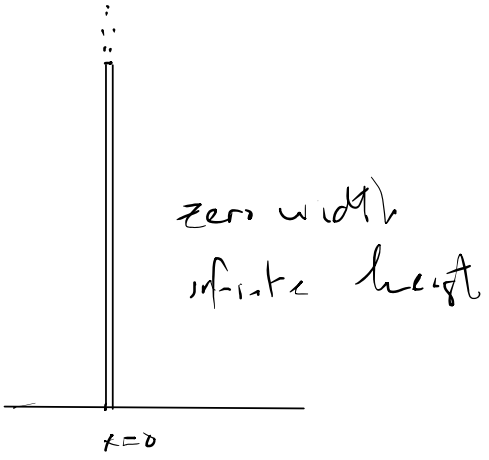
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$$\delta(x) = 0, \quad x \neq 0$$

$$f(0) = \int_a^b f(x)\,\delta(x)\,dx$$

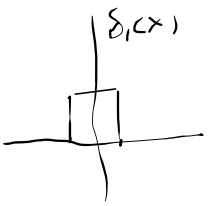
$$\int_{-\infty}^{\infty} \delta(x)\,dx = 1$$

$a < 0 < b$



$\delta(x)$ as a limit of $\delta_n(x)$ sequences:

Ex $\delta_n(x) = \begin{cases} 0 & ; \quad x < \frac{-1}{2n} \\ n & ; \quad \frac{-1}{2n} < x < \frac{1}{2n} \\ 0 & ; \quad x > \frac{1}{2n} \end{cases}$



Ex $\delta_n(x) = \frac{n}{\sqrt{\pi}} e^{\frac{x^2}{1/n^2}}$

$\delta_n(x) = \frac{n}{\pi} \frac{1}{1+n^2x^2}$

$\delta_n(x) = \frac{\sin nx}{\pi x} = \frac{1}{2\pi} \int_{-n}^{n} e^{ixt}\, dt$

Distribution

$$\int_{-\infty}^{\infty} \delta(x) f(x)\,dx = \lim_{n\to\infty} \int_{-\infty}^{\infty} \delta_n(x) f(x)\,dx$$

↓

Distrib.

Properties

Even

$$\delta(x) = \delta(-x)$$

Scaling

$$\delta(ax) = \frac{1}{a}\delta(x)$$

proof: $f(0) = \int_{-\infty}^{\infty} f(x)\,\delta(ax)\,dx$

$y = ax \Rightarrow \int_{-\infty}^{\infty} f\left(\frac{y}{a}\right) \frac{\delta(y)}{a}\,dy$

Shift of origin

$$\int_{-\infty}^{\infty} f(x)\ \delta(x - x_0)\, dx = f(x_0)$$

Composition

$$\delta(g(x)) = \sum_i \frac{\delta(x - a_i)}{|g'(a_i)|}$$

$a_i \equiv$ zeros of $g(x)$

$g'(a_i) \neq 0$

Proof

$$\int_{-\infty}^{\infty} f(x)\, \delta(g(x))\, dx = \sum_i \int_{a_i - \varepsilon}^{a_i + \varepsilon} f(x)\, \delta\left[(x - a_i)\, g'(a_i)\right] dx$$

$$= \sum_i \int_{a_i - \varepsilon}^{a_i + \varepsilon} f(x)\, \frac{\delta(x - a_i)}{|g'(a_i)|}\, dx = \sum_i \frac{f(a_i)}{|g'(a_i)|}$$

$$g(x) = \cancel{g(a_i)} + (x - a_i)\, g'(a_i) + \cdots$$

Ex $\delta[(x-a)(x-b)] = \frac{1}{|a-b|}\left(\delta(x-a)+\delta(x-b)\right)$

$\delta(x^2-a^2) = \frac{1}{2|a|}\left[\delta(x-a)+\delta(x+a)\right]$

Derivative

$$\int_{-\infty}^{\infty} f(x)\,\delta'(x-x_0)\,dx = -\int_{-\infty}^{\infty} f'(x)\,\delta(x-x_0)\,dx = -f'(x_0)$$

$$\int_{-\infty}^{\infty} f(x)\,\delta^{(n)}(x-x_0)\,dx = (-1)^n f^{(n)}(x_0)$$

Ind. Integral



$$\int f(x)\,\delta(x-x_0)\,dx = f(x_0)\,u(x-x_0) + C$$

$$\int \delta(x-x_0)\,dx = u(x-x_0) + C$$

Heaviside step function

3D

$$\int f(\vec{r}')\,\delta(\vec{r}-\vec{r}')\,d^3\vec{r}' = f(\vec{r})$$

Ch. 5

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Vector Space

$\vec{v}$ → add $\vec{v} \in V$

→ mult. scalar $c \in F$

Ex $\mathbb{R}^n$

$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2$

$\vec{v} + \vec{u} = \begin{pmatrix} v_1 + u_1 \\ v_2 + u_2 \end{pmatrix}$

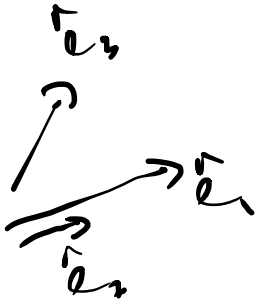
$c\vec{v} = \begin{pmatrix} cv_1 \\ cv_2 \end{pmatrix}$

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = v_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + v_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Ex $\mathbb{R}^n$ $\qquad \vec{v} = c_1 \hat{e}_1 + c_2 \hat{e}_2 + \cdots + c_n \hat{e}_n$

$\{\hat{e}_i\}$ basis

Ex



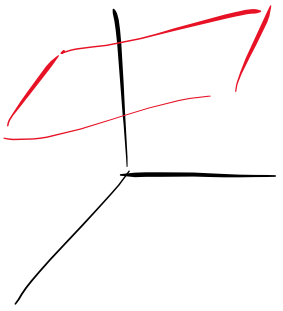
Subspace

Ex $\mathbb{R}^3$

$W \subset V$

1) $w_1 + w_2 \in W$

2) $cw \in W$



Ex $x + 2y + z = 3$

$x_1 + 2y_1 + z_1 = 3$

$x_2 + 2y_2 + z_2 = 3$ $+ \Rightarrow (x_1 + x_2) + 2(y_1 + y_2) + (z_1 + z_2) = 6$

Linear Independence

$\mathbb{R}^1$ $\quad \vec{v} = c\vec{b}$



$\mathbb{R}^2$ $\quad \vec{v}_1 = c\vec{v}_2 \quad \Rightarrow \quad c_1\vec{v}_1 + c_2\vec{v} = 0$

$\mathbb{R}^3$ $\quad \vec{v}_3 = c_1\vec{v}_1 + c_2\vec{v}_2 \quad\quad c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v} = 0$

Basis and vectors

$\vec{v} \in \mathbb{R}^3 \qquad \text{basis} = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\}$

$$\vec{v} = v_1\vec{b}_1 + v_2\vec{b}_2 + v_3\vec{b}_3 = \begin{bmatrix} | & | & | \\ \vec{b}_1 & \vec{b}_2 & \vec{b}_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = B\,v$$

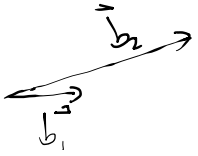
Standard basis

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

Different basis

$$\vec{V} = \begin{bmatrix} \vec{b}_1' & \vec{b}_2' & \vec{b}_3' \end{bmatrix} \begin{bmatrix} v_1' \\ v_2' \\ v_3' \end{bmatrix}$$

Choice of basis





Scalar Product

$$\vec{v}\cdot\vec{u} = v_1u_1+v_2u_2+\cdots = v^T u = \begin{pmatrix} v_1 & v_2 & \cdots \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \\ \vdots \end{pmatrix}$$

$\Rightarrow$ Distance, angles,

$$|\vec{a}\cdot\vec{b}|^2 \leq |\vec{a}\cdot\vec{a}|\,|\vec{b}\cdot\vec{b}|$$

Orthogonal basis

$\vec{b}_i\cdot\vec{b}_j = 0 \quad i\neq j \quad \Rightarrow \quad \vec{v} = \sum_i v_i\vec{b}_i$

$$\vec{b}_j\cdot\vec{v} = \sum_i v_i\vec{b}_i\cdot\vec{b}_j$$

$$\vec{v}\cdot\vec{b}_j = v_j(\vec{b}_j\cdot\vec{b}_j)$$

$$v_j = \frac{\vec{v}\cdot\vec{b}_j}{|\vec{b}_j|^2}$$

Orthonormal Basis

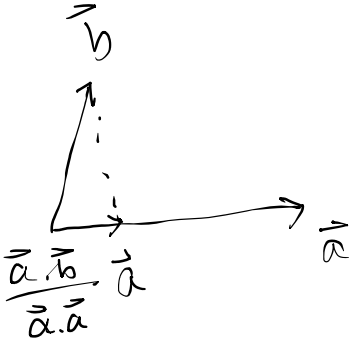
$$\vec{b}_i \cdot \vec{b}_j = \delta_{ij}$$

$$v_i = \vec{v} \cdot \vec{b}_i$$

Gram-Schmit Orthogonalization

$\vec{a}$, $\vec{b}$

$$\vec{a}_1 = a \qquad \vec{b}_1 = \vec{b} - \frac{\vec{a}\cdot\vec{b}}{\vec{a}\cdot\vec{a}}\vec{a}$$



Ex $\vec{a} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $\vec{b} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$

$\vec{a}_1 = \vec{a}$

$\vec{b}_1 = \vec{b} - \frac{\vec{a}\cdot\vec{b}}{\vec{a}\cdot\vec{a}}\vec{a} = \begin{bmatrix} 2 \\ 6 \end{bmatrix} - \frac{10}{5}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$

$\vec{a}_1 \cdot \vec{b}_1 = 0$

normalization $\hat{a}_1 = \frac{\vec{a}_1}{\sqrt{\vec{a}_1\cdot\vec{a}_1}} = \frac{1}{\sqrt{5}}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $\hat{b}_1 = \frac{\vec{b}_1}{\sqrt{\vec{b}_1\cdot\vec{b}_1}} = \frac{1}{\sqrt{18}}\begin{bmatrix} -2 \\ 4 \end{bmatrix}$

Three vectors $\vec{a}, \vec{b}, \vec{c}$

$\vec{a}_1 = \vec{a}$ $\vec{b}_1 = \vec{b} - \frac{\vec{a}\cdot\vec{b}}{\vec{a}\cdot\vec{a}}\vec{a}$

$\vec{c}_1 = \vec{c} - \frac{\vec{c}\cdot\vec{a}}{\vec{a}\cdot\vec{a}}\vec{a} - \frac{\vec{c}\cdot\vec{b}}{\vec{b}\cdot\vec{b}}\vec{b}$

Ex Change of basis

$$\vec{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \Rightarrow \quad \vec{v} \cdot \vec{v} = 13$$

$$\vec{v} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 4 \end{bmatrix} = -1\,\vec{b}_1 + 4\,\vec{b}_2$$

$$\vec{v} \cdot \vec{v} = \left(-1\,\vec{b}_1 + 4\,\vec{b}_2\right) \cdot \left(-1\,\vec{b}_1 + 4\,\vec{b}_2\right) = (-1)(-1)\,\vec{b}_1 \cdot \vec{b}_1 + (-1)4\,\vec{b}_1 \cdot \vec{b}_2 + (4)(-1)\,\vec{b}_2 \cdot \vec{b}_1 + 4(4)\,\vec{b}_2 \cdot \vec{b}_2$$

$$= 5 - 4(3) - 4(3) + 16(2) = 13$$

$$\vec{v}\cdot\vec{w} = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} \begin{bmatrix} \vec{b}_1\cdot\vec{b}_1 & \vec{b}_1\cdot\vec{b} & \cdots \\ \vec{b}_2\cdot\vec{b}_1 & \ddots & \\ 1 & & \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$$

$$= v^T B w$$

↘

"Dot product matrix"

"Gramian"

"Metric"

Transformations

vector space

$T(\vec{v}) = \vec{V}$

$T : V \rightarrow V$

Linear transformations

1) $T(\vec{v} + \vec{w}) = T\vec{v} + T\vec{w}$

2) $T(c\vec{v}) = cT\vec{v}$

$\hookrightarrow T(a_1\vec{v}_1 + a_2\vec{v}_2) = a_1 T\vec{v}_1 + a_2 T\vec{v}_2$

Matrix Representation

Action on basis

$$T\hat{e}_1 = T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}a\\b\end{bmatrix}$$

$$T\hat{e}_2 = T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}c\\d\end{bmatrix}$$

$$T\vec{v} = T(v_1\hat{e}_1 + v_2\hat{e}_2)$$

$$= v_1 T\hat{e}_1 + v_2 T\hat{e}_2 = v_1\begin{bmatrix}a\\b\end{bmatrix} + v_2\begin{bmatrix}c\\d\end{bmatrix}$$

$$= \begin{bmatrix}a & c\\ b & d\end{bmatrix}\begin{bmatrix}v_1\\v_2\end{bmatrix}$$

$$\downarrow$$

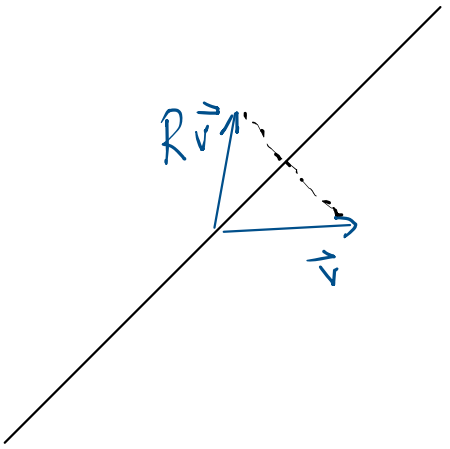
$$M(T)$$

Transformations

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Reflection in line

R



Eigen vector

$R\vec{v} = \lambda\vec{v}$

$R^2 = I \Rightarrow R^2 - I = 0$

$x^2 - 1 = 0$

$\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$

Projection

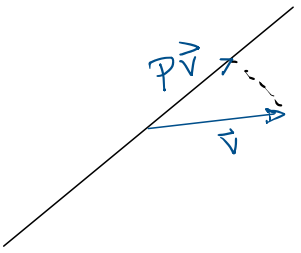
P

$P^2 - P = 0$

$x^2 - x = 0$

$\lambda^2 - \lambda = 0$

$\lambda = 0, 1$



Rotation in Plane

R_θ

$R_\pi^2 = I$

Translation

$S\vec{v} = \vec{v} + \vec{s}$

$S(\vec{v} + \vec{w}) = \vec{v} + \vec{w} + \vec{s}$

$\neq S(\vec{v}) + S(\vec{w})$

Eigenvalue Problem Importance

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* Identity of Transformation
* Diagonalization
* Functions of Tranformations

$$f\left(\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}\right) = \begin{bmatrix} f(a) & 0 \\ 0 & f(b) \end{bmatrix}$$

Ex $A = \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix}$ $A\vec{v} = \lambda\vec{v} \Rightarrow (A - \lambda I)\vec{v} = 0$

$|A - \lambda I| = 0$ $\begin{vmatrix} 5-\lambda & -2 \\ -2 & 2-\lambda \end{vmatrix} = 0$

$(5-\lambda)(2-\lambda) - 4 = 0 \Rightarrow \lambda = 1, 6$

$\lambda_1 = 1$ $\begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $x_2 = 2x_1 \Rightarrow \vec{v}_1 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$\lambda_2 = 6$ $\begin{pmatrix} -1 & -2 \\ -2 & -4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $x_1 = -2x_2 \Rightarrow \vec{v}_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$$A\begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix} = P \begin{pmatrix} 1 & 0 \\ 0 & 6 \end{pmatrix}$$

(P over the first matrix, Λ over the second)

$$AP = P\Lambda$$

Symmetric: $A = A^T$

1) Real numbers eigenvalues

2) Orthogonal eigenvectors

$$A = Q\Lambda Q^T$$

$$P^T = \begin{bmatrix} - & \vec{v}_1 & - \\ - & \vec{v}_2 & - \end{bmatrix}$$

$$P = \begin{bmatrix} | & | \\ \vec{v}_1 & \vec{v}_2 \\ | & | \end{bmatrix}$$

$$P^T P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$P^T = P^{-1}$$

$P \equiv$ Orthogonal

Symmetric Matrix

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$A = A^T$ *

$A = Q \Lambda Q^T$

Proof

① $Av = \lambda v \Rightarrow A v^* = \lambda^* v^* \Rightarrow$ (transpose) $\left(v^{*T} A = \lambda^* v^{*T} \right) v$ (* on A)

$v^{*T} A v = \lambda^* |v|^2$

$\lambda |v|^2 = \lambda^* |v^2| \Rightarrow \lambda = \lambda^*$

② $Av = \lambda_v v \qquad Aw = \lambda_w w$

$(Av)^T w = (A^T v)^T w = \boxed{v^T A w} = \lambda_w v^T w = \lambda_w \vec{v} \cdot \vec{w}$ (* on Av)

$\lambda_v v^T w \Rightarrow \lambda_v \vec{v} \cdot w = \lambda_w \vec{v} \cdot \vec{w}$

If $\lambda_v \neq \lambda_w \Rightarrow \vec{v} \cdot \vec{w} = 0$

Orthogonal Matrix

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$$Q^{-1} = Q^T \Rightarrow Q^T Q = I \Rightarrow \det(Q)^2 = 1$$

$\det(Q) = +1 \Rightarrow$ rotation

$\det(Q) = -1 \Rightarrow$ rotation + reflection

$$Qr = R$$

$$\Rightarrow R^T R = (Qr)^T (Qr) = r^T Q^T Q r$$

$$= r^T r$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$



Ex $B = \frac{1}{3}\begin{bmatrix} -2 & -1 & -2 \\ 2 & -2 & -1 \\ 1 & 2 & -2 \end{bmatrix}$ $\det(B) = -1$

$\lambda = -1$ $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ Reflection in plane $\perp \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$x - y + z = 0$

$\lambda = 1$ any two vectors in $(x - y + z = 0)$ plane

$U = \begin{bmatrix} 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ -2/\sqrt{6} & 0 & 1/\sqrt{3} \end{bmatrix} \Rightarrow U^{-1}BU = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$\theta = 120°$ $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Similarity Transf.

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~~$r' = P r$~~

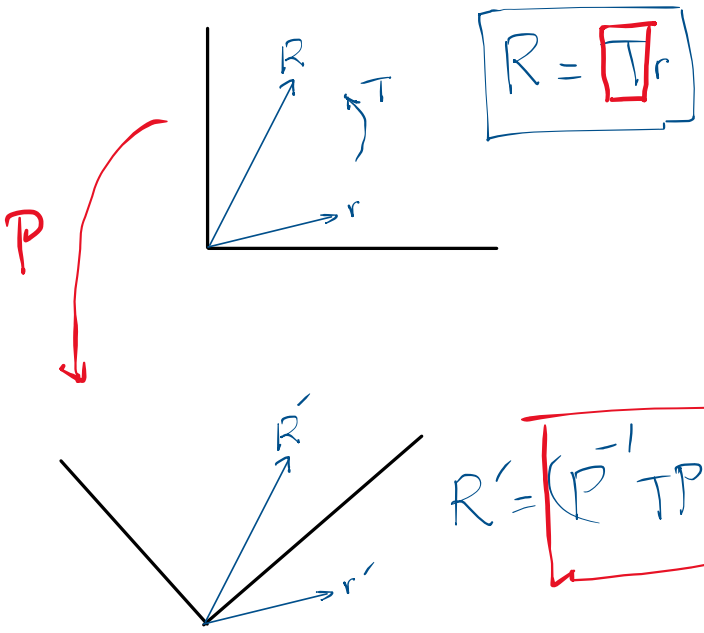
$r = P r'$

$R = P R'$

$T r = R = P R'$

$T P r' = P R'$

$R' = (P^{-1} T P) r'$



$R = T r$

$R' = (P^{-1} T P) r'$

$T \rightarrow P^{-1} T P$

General Vector Spaces

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Ex M_{22} $\quad A \in M_{22} \Rightarrow A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Basis $E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \ldots,$ --

$$A = a E_{11} + b E_{12} + c E_{21} + d E_{22}$$

Ex $\mathbb{C} \subset M_{22}$ $\quad \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$

$$Re = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad Im = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = a\,Re + b\,Im$$

Function Spaces

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$(f+g)(x) = f(x) + g(x)$

$c\,f(x) = (cf)(x)$

Ex $1, \cos^2 x, \sin^2 x$ linearly dependent

Ex P_n Basis $= \{1, x, x^2, \dots, x^n\}$

$n+1$

Ex $ay'' + by' + cy = 0 \Rightarrow$ solution space

$y'' = 0 \Rightarrow y(x) = A + Bx$

5.1 Vectors in Function Spaces

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Ex $P_0(s) = 1$, $P_1(s) = s$, $P_2(s) = \frac{3}{2}s^2 - \frac{1}{2}$

$s + 3 = 3P_0 + P_1$ $\qquad s^2 = \frac{1}{3}P_0 + \frac{2}{3}P_2$

Ex $\varphi_n(s) = s^n$; $n = 0, 1, 2, \cdots$ $\Rightarrow$ Maclaurin Series

$$f(s) = a_0 + a_1 s + a_2 s^2 + \cdots = \sum_{n=0}^{\infty} a_n s^n$$

Ex Span Space

$f = a_1\alpha + a_2\beta$

$g = b_1\alpha + b_2\beta$

$f + ig = (a_1 + ib_1)\alpha + (a_2 + ib_2)\beta$

Scalar Product

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$[f,g]$ (f,g)

$$\langle f|g\rangle = \int_a^b f^*(s)\, g(s)\, w(s)\, ds$$

$$\vec{V} = \sum v_i \hat{e}_i$$

$$f(x) = \int dx\, f_i(x_i)\, \delta(x - x_i)$$

2x $\langle\alpha|\alpha\rangle = 1 = \langle\beta|\beta\rangle$ $\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle = 0$

$f = a_1\alpha + a_2\beta$

$g = b_1\alpha + b_2\beta$

$$\langle f|g\rangle = \langle a_1\alpha + a_2\beta \,|\, (b_1\alpha + b_2\beta)\rangle$$

$$= a_1 b_1 \langle\alpha|\alpha\rangle + a_1 b_2 \langle\alpha|\beta\rangle + a_2 b_1 \langle\beta|\alpha\rangle + a_2 b_2 \langle\beta|\beta\rangle$$

$$= a_1 b_1 + a_2 b_2$$

Hilbert Space ($\mathcal{H}$)

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Elements: functions; f, g, or h $\in \mathcal{H}$

Basis: $\{\varphi_i\} \Rightarrow f(s) = \sum_{n=0}^{\infty} a_n \varphi_n(s)$

Completeness

Scalar: $\mathbb{C}$

Scalar Product: $\langle f|g\rangle$

1) $\langle f|f\rangle \geq 0$ "0 when $f = 0$" $\Rightarrow$ $\langle f|f\rangle^{1/2} \equiv$ norm of f

2) $\langle g|f\rangle^{*} = \langle f|g\rangle$

$\langle f|g+h\rangle = \langle f|g\rangle + \langle f|h\rangle$; $\langle f|kg\rangle = k\langle f|g\rangle$

Ex $p(x) = 1 + x$

$q(x) = 1 - x$

$$\langle p|q \rangle = \int_0^1 p(x)\, q(x)\, x\, dx$$

$$= \int_0^1 x(x+1)(1-x)\, dx = \frac{1}{4}$$

$\text{Basis} = \{1, 1+x, x^2\} \Rightarrow B = \begin{bmatrix} 1/2 & 5/6 & 1/4 \\ 5/6 & 17/12 & 9/20 \\ 1/4 & 9/20 & 1/6 \end{bmatrix}$

"Dot product matrix"

$p(x) \longrightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$q(x) \longrightarrow \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$

$$\langle p|q \rangle = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} B \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = 2\left(\frac{5}{6}\right) - 1\left(\frac{17}{12}\right) = \frac{1}{4}$$

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Schwarz Inequality

$$|\langle f|g\rangle|^2 \leq \langle f|f\rangle\langle g|g\rangle$$

Orthonormal Basis

$$\{\phi_i\} \Rightarrow f = \sum_i a_i \phi_i \Rightarrow \langle \phi_j | f\rangle = \sum_i a_i \langle \phi_j|\phi_i\rangle$$

$$\langle \phi_j|\phi_i\rangle = \delta_{ij} \Rightarrow \boxed{a_j = \langle \phi_j|f\rangle}$$

Ex 5.1.3

$\chi_n(x) = \sin nx$, $n = 1, 2, \cdots$ $\quad 0 \leq x \leq \pi$

$\phi_n(x) = \sqrt{\frac{2}{\pi}} \sin nx$

$$\langle \chi_n|\chi_m\rangle = \int_0^\pi \sin(nx)\sin(mx)\,dx = \begin{cases} 0 & n \neq m \\ \frac{\pi}{2} & n = m \end{cases}$$

Expansion of Scalar Product

$$f = \sum_\mu a_\mu \varphi_\mu \qquad g = \sum_\nu b_\nu \varphi_\nu$$

$$\langle f|g\rangle = \sum_{\mu\nu} a_\mu^* b_\nu \langle \varphi_\mu|\varphi_\nu\rangle = \sum_\mu a_\mu^* b_\mu \Rightarrow a^\dagger b$$

$$\langle f|f\rangle = \sum_\mu |a_\mu|^2 \geq 0 \Rightarrow a^\dagger a$$

Expansion of δ-function

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$$\delta(x-t) = \sum_{n=0}^{\infty} a_n(t)\, \varphi_n(x) \quad \Rightarrow \quad a_n(t) = \int_a^b \varphi_n^*(x)\, \delta(x-t)\, dx$$

$$= \varphi_n^*(t)$$

$$\delta(x-t) = \sum_{n=0}^{\infty} \varphi_n^*(t)\, \varphi_n(x)$$

Ex $\varphi_n(x) = \sqrt{2}\, \sin(n\pi x)$; $n = 1, 2, 3, \ldots$ on $(0,1)$

$$\delta(x-t) = \lim_{N\to\infty} 2 \sum_{n=0}^{N} \sin(n\pi t)\, \sin(n\pi x)$$

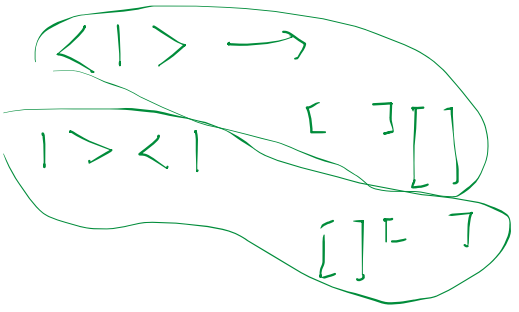
Dirac Notation

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Kets $f \rightarrow |f\rangle$

$\varphi_i \rightarrow |\varphi_i\rangle$

bras $\varphi_i^* \rightarrow \langle\varphi_i|$



$$|f\rangle = \sum_i a_i |\varphi_i\rangle = \sum_i |\varphi_i\rangle\langle\varphi_i|f\rangle = \left(\sum_i |\varphi_i\rangle\langle\varphi_i|\right)|f\rangle$$

$$I = \sum_i |\varphi_i\rangle\langle\varphi_i|$$

Resolution of Identity

Gram–Schmidt Orth.

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$\chi_\mu \rightarrow \varphi_\nu$ orthonormalize sequentially

$$\varphi_0 = \frac{\chi_0}{\langle \chi_0 | \chi_0 \rangle^{1/2}}$$

$$a_{1,0} = \langle \varphi_0 | \chi_1 \rangle$$

$$\psi_1 = \chi_1 - a_{1,0}\varphi_0 \Rightarrow 0 = \langle \varphi_0 | \psi_1 \rangle = \langle \varphi_0 | \chi_1 \rangle - a_{1,0} \Rightarrow$$

$$\psi_1 = \chi_1 - \langle \varphi_0 | \chi_1 \rangle \varphi_0 \Rightarrow \varphi_1 = \frac{\psi_1}{\langle \psi_1 | \psi_1 \rangle^{1/2}}$$

$$\psi_2 = \chi_2 - \langle \varphi_0 | \chi_2 \rangle \varphi_0 - \langle \varphi_1 | \chi_2 \rangle \varphi_1 \Rightarrow \varphi_2 = \frac{\psi_2}{\langle \psi_2 | \psi_2 \rangle^{1/2}}$$

$$\psi_n = x_n - \sum_{i=0}^{n-1} \langle \varphi_i | x_n \rangle \varphi_i$$

$$\varphi_n = \frac{\psi_n}{\langle \psi_n | \psi_n \rangle^{1/2}}$$

Ex x_μ as x^μ $\qquad \langle f|g \rangle = \int_{-1}^{1} f^* g \, dx$

$$\varphi_0(x) = \frac{1}{\langle 1|1 \rangle^{1/2}} = \frac{1}{\sqrt{2}}$$

$$\psi_1(x) = x - \langle x | \tfrac{1}{\sqrt{2}} \rangle \tfrac{1}{\sqrt{2}} = x \quad \Rightarrow \varphi_1(x) = \frac{x}{\langle x|x \rangle^{1/2}} = \sqrt{\tfrac{3}{2}}\, x$$

$$\psi_2(x) = x^2 - \langle x^2 | \tfrac{\sqrt{3}}{2} x \rangle \tfrac{\sqrt{3}}{2} x - \langle x^2 | \tfrac{1}{\sqrt{2}} \rangle \tfrac{1}{\sqrt{2}} \quad \Rightarrow \varphi_2(x) = \frac{\sqrt{5}}{2}\left(\tfrac{3}{2}x^2 - \tfrac{1}{2}\right)$$

$$= x^2 - \tfrac{1}{3}$$

$$\phi_3(x) = \sqrt{\frac{7}{2}}\left(\frac{5}{2}x^3 - \frac{3}{2}x\right)$$

$$\phi_n(x) = \sqrt{\frac{2n+1}{2}}\,P_n(x)$$

5.3 Operators

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Operator: $\mathcal{H} \to \mathcal{H}$

Ex
* $c, x,$
* $\frac{d}{dx}$
* $\int G(x,x') f(x') dx'$

Linear

Differential Operator

$$\mathcal{L} = (1-x^2)\frac{d^2}{dx^2} - 2x\frac{d}{dx}$$

$$\mathcal{L}y = -\lambda y$$

Commutation of operators

$$[A, B] = AB - BA$$

$p = -i\frac{d}{dx} \Rightarrow [x,p]f(x) = -ix\frac{d}{dx}f + i\frac{d}{dx}(xf)$

$= +if$

$\Rightarrow [x,p] = i$

Properties

$[A,B] = -[B,A]$

$[A,B+C] = [A,B] + [A,C]$

$k[A,B] = [kA,B] = [A,kB]$

Ex $[x,p^2] = [x,p]p + p[x,p] = 2ip$

Special Operators

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Identity & Inverse

$$A^{-1}A = AA^{-1} = I$$

Adjoint

$\langle f|Ag\rangle = \langle A^\dagger f|g\rangle$ $\quad A^\dagger \equiv$ adjoint of A

$H^\dagger = H \Rightarrow H \equiv$ Hermitian

$U^\dagger = U^{-1} \Rightarrow U \equiv$ unitary

Ex $\quad A = \frac{d}{dx}$ $\quad \langle f|g\rangle = \int_{-\infty}^{\infty} f^*(x) g(x)\, dx$ $\quad L^2$ on $(-\infty, \infty)$

$$\langle f|Ag\rangle = \int_{-\infty}^{\infty} f^* \frac{dg}{dx}\, dx = f^* g\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{df^*}{dx}\, g\, dx$$

$$= \int_{-\infty}^{\infty} \left(-\frac{d}{dx} f\right)^* g\, dx \Rightarrow A^\dagger = -\frac{d}{dx}$$

Ex $A = i\frac{d}{dx}$ $\quad \langle f|g\rangle = \int_{-\infty}^{\infty} f^*(x) g(x)\, dx$ $\quad L^2$ on $(-\infty, \infty)$

$$\langle f|Ag\rangle = \int_{-\infty}^{\infty} f^* i\frac{dg}{dx}\, dx = f^* g\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{df^*}{dx} i g\, dx$$

$$= \int_{-\infty}^{\infty} -\left(-i\frac{d}{dx} f\right)^* g\, dx \Rightarrow A^\dagger = i\frac{d}{dx}$$

Ex $A = -i\frac{d}{dx}$ in L^2 on $(-1, 1)$

not Hermitian

$$\langle f|Ag\rangle = -i f^* g\Big|_{-1}^{1} - \int_{-1}^{1} \frac{df^*}{dx} (-i g)\, dx$$

Basis expansion

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$\chi = \sum b_\nu \varphi_\nu$

$$A\varphi_\mu = \chi = \sum_\nu a_{\nu\mu}\,\varphi_\nu \quad \Rightarrow \quad a_{\nu\mu} = \langle \varphi_\nu | A \varphi_\mu \rangle = \langle \varphi_\nu | A | \varphi_\mu \rangle$$

$$\psi = \sum_\mu c_\mu \varphi_\mu \Rightarrow \quad A\psi = \sum_\mu c_\mu A\varphi_\mu = \sum_\mu c_\mu \sum_\nu a_{\nu\mu}\varphi_\nu$$

$$= \sum_\nu \underbrace{\left(\sum_\mu a_{\nu\mu} c_\mu\right)}_{b_\nu} \varphi_\nu$$

$\psi \to c$

$\chi \to b$

$A \to \mathbb{A}$

$$\Rightarrow \quad \vec{b} = \mathbb{A}\,\vec{c}$$

Dirac Notation

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$$A\psi = \left(\sum_{\nu\mu} |\varphi_\nu\rangle\langle\varphi_\nu|A|\varphi_\mu\rangle\langle\varphi_\mu|\right)|\psi\rangle$$

$$A\,|\psi\rangle$$

$$\left(\sum_\nu |\varphi_\nu\rangle\langle\varphi_\nu|\right)A\left(\sum_\mu |\varphi_\mu\rangle\langle\varphi_\mu|\right)|\psi\rangle$$

$$A = \sum_{\nu\mu} |\varphi_\nu\rangle\, a_{\nu\mu}\, \langle\varphi_\mu|$$

ex $\quad I = \sum_{\nu\mu} |\varphi_\nu\rangle\,\delta_{\nu\mu}\,\langle\varphi_\mu| = \sum_\mu |\varphi_\mu\rangle\langle\varphi_\mu|$

Expansion of $A^\dagger$

$$\langle x|A^\dagger|\psi\rangle = \langle Ax|\psi\rangle = \langle\psi|A|x\rangle^* = \left[\langle\psi|\left(\sum_{\nu\mu}|\varphi_\nu\rangle\, a_{\nu\mu}\,\langle\varphi_\mu|\right)|x\rangle\right]^*$$

$$= \sum_{\nu\mu} \langle\psi|\varphi_\nu\rangle^*\, a^*_{\nu\mu}\, \langle\varphi_\mu|x\rangle^* = \sum_{\nu\mu}\langle x|\varphi_\mu\rangle\, a^*_{\nu\mu}\,\langle\varphi_\nu|\psi\rangle$$

$$A^\dagger = \sum_{\nu\mu} |\varphi_\mu\rangle\, a^*_{\nu\mu}\,\langle\varphi_\nu|$$

Example

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Ex 5.3.4

x in term of $\varphi_n(x) = c_n H_n(x) e^{-x^2/2}$; $n = 0, 1, 2, \ldots$

$$\langle f | g \rangle = \int_{-\infty}^{\infty} f^*(x) g(x) dx$$

$$x_{\nu\mu} = \langle \varphi_\nu | x | \varphi_\mu \rangle = c_\nu c_\mu \int_{-\infty}^{\infty} H_\nu(x) H_\mu(x) e^{-x^2} dx$$

$$x = \begin{bmatrix} 0 & \sqrt{1}/2 & 0 & 0 & \cdots \\ \sqrt{1}/2 & 0 & 1 & 0 & \cdots \\ 0 & 1 & 0 & \sqrt{6}/2 & \cdots \\ 0 & 0 & \sqrt{6}/2 & 0 & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$$

5.4 Self-adjoint Operator

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Self-adjoint "Hermitian" $\longrightarrow$ Observable $\qquad \langle X \rangle = \int x \rho(x) dx$

Expectation values: $\langle A \rangle = \langle \psi | A | \psi \rangle = \langle A^\dagger \psi | \psi \rangle$

$\langle A \rangle^* = \langle A\psi | \psi \rangle$

$A = A^\dagger$ self-adjoint

$a_{\nu\mu} = a^*_{\mu\nu}$

$\langle A \rangle \Rightarrow$ real

$\langle A \rangle$ basis expansion

$$\langle A \rangle = \langle \psi | A | \psi \rangle = \left\langle \sum_\nu c_\nu \psi_\nu \middle| A \middle| \sum_\mu c_\mu \psi_\mu \right\rangle = \sum_{\nu\mu} c_\nu^* a_{\nu\mu} c_\mu = c^\dagger A c$$

$(c^\dagger A c)^\dagger = c^\dagger A c \Rightarrow$ real

5.5 Unitary Operators

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Transformation between orthonormal bases:

$$\psi = \sum_\mu c_\mu \varphi_\mu = \left(\sum_\mu |\varphi_\mu\rangle\langle\varphi_\mu|\right)|\psi\rangle$$

$$V = U^{-1} = U^\dagger$$

Basis transf.

$$\varphi_\mu = \sum_\nu u_{\nu\mu}\varphi'_\nu = \left(\sum_\nu |\varphi'_\nu\rangle\langle\varphi'_\nu|\right)|\varphi_\mu\rangle \Rightarrow \boxed{u_{\nu\mu} = \langle\varphi'_\nu|\varphi_\mu\rangle}$$

$$\psi = \sum_\mu c_\mu \sum_\nu u_{\nu\mu}\varphi'_\nu = \sum_\nu \left(\sum_\mu u_{\nu\mu} c_\mu\right)\varphi'_\nu = \sum_\nu c'_\nu \varphi'_\nu \Rightarrow \boxed{c' = Uc}$$

Reverse Trans

$$\varphi'_\mu = \sum_\nu v_{\nu\mu}\varphi_\nu = \sum_\nu \langle\varphi_\nu|\varphi'_\mu\rangle\varphi_\nu \Rightarrow v_{\nu\mu} = \langle\varphi_\nu|\varphi'_\mu\rangle = (U^*)_{\mu\nu}$$

$$= U^\dagger_{\nu\mu}$$

Unitary Transformations

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Unitary Trans.: $\{\phi_\mu\} \longrightarrow \{\phi'_\nu\}$

$c \qquad\qquad c'$

$$\Rightarrow c' = Uc \qquad\qquad U^\dagger U = I$$

$$U_{\nu\mu} = \langle \varphi'_\nu | \varphi_\mu \rangle$$

Transform. of Operators

$$A' = UAU^\dagger = UAU^{-1}$$

$$Ac = b \Rightarrow A'c' = b'$$

Invariants

$f \to a \qquad f = \sum a_\mu \varphi_\mu \qquad g \to b$

$$\Rightarrow \langle f|g\rangle = (a')^\dagger b' = (Ua)^\dagger Ub = a^\dagger U^\dagger U b = a^\dagger b$$

Expectation value

$$\psi = \sum_i c_i \varphi_i$$

$$\langle A \rangle = \langle \psi | A | \psi \rangle = c^\dagger A c$$

$$c^\dagger A' c' =$$

$$(Uc)^\dagger (UAU^\dagger)(Uc) = c^\dagger U^\dagger U A U^\dagger U c = c^\dagger A c$$

Determinant

Trace

$$\text{trace}(A') = \sum_\nu (UAU^{-1})_{\nu\nu} = \sum_{\nu\mu\tau} U_{\nu\mu} a_{\mu\tau} (U^{-1})_{\tau\nu} = \sum_{\mu\tau} \left(\sum_\nu (U^{-1})_{\tau\nu} U_{\nu\mu} \right) a_{\mu\tau}$$

$$= \sum_{\mu\tau} \delta_{\mu\tau} a_{\mu\tau} = \sum_\mu a_{\mu\mu} = \text{trace}(A)$$

Ch.6 Eigenvalue Problem

Wednesday, February 25, 2026 3:16 PM

Eigenvalue Eqn

$A\psi = \lambda \psi$ → eigenfunction

↓ eigenvalue

Physics examples

1) $A\psi \Rightarrow$ force $\Rightarrow$ $A\psi = \lambda\psi \Rightarrow$ standing waves

$\psi \Rightarrow$ displacement

2) $L = I\omega \Rightarrow I\omega = \lambda\omega \Rightarrow$ principal axes

3) $H\psi = \lambda\psi \Rightarrow \lambda = E, \psi$

Basis expansion

$a_{ij} = \langle \varphi_i | A | \varphi_j \rangle \qquad c_i = \langle \varphi_i | \psi \rangle$

$Ac = \lambda c \Rightarrow \psi = \sum_i c_i \varphi_i$

6.2 Matrix Eigenvalue Problem

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2D motion in ellipsoidal basin

$$V(x,y) = ax^2 + bxy + cy^2$$

$$F_x = -\frac{\partial V}{\partial x} = -2ax - by$$

$$F_y = -\frac{\partial V}{\partial y} = -bx - 2cy$$

$$\begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} -2a & -b \\ -b & -2c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$f = H r \quad \Rightarrow \quad \underline{Hr = \lambda r}$$

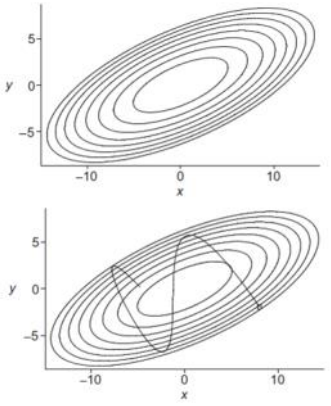


FIGURE 6.1 Top: Contour lines of basin potential $V = x^2 - \sqrt{5}xy + 3y^2$. Bottom: Trajectory of sliding particle of unit mass starting from rest at $(8.0, -1.92)$.

Ex 6.2.1

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$a=1,\quad b=-\sqrt{5},\quad c=3$

$$H=\begin{pmatrix}-2 & \sqrt{5}\\ \sqrt{5} & -6\end{pmatrix} \Rightarrow \det(H-\lambda I)=\lambda^2+8\lambda+7=0 \Rightarrow \lambda=-1,-7$$

$$\lambda_1=-1,\ \vec{r}_1=c\begin{bmatrix}\sqrt{5}\\ 1\end{bmatrix}=\begin{bmatrix}\sqrt{5/6}\\ 1/\sqrt{6}\end{bmatrix}$$

$$\lambda_2=-7,\ \vec{r}_2=c'\begin{bmatrix}-1\\ \sqrt{5}\end{bmatrix}=\begin{bmatrix}-1/\sqrt{6}\\ \sqrt{5/6}\end{bmatrix}$$

} Pricipal axes

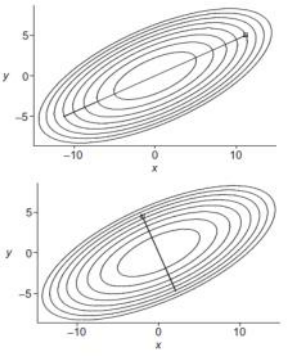


FIGURE 6.2 Trajectories starting at rest. Top: At a point on the line $x = y\sqrt{5}$. Bottom: At a point on the line $y = -x\sqrt{5}$.

$$\vec{U}\vec{r}=\begin{bmatrix}\sqrt{5/6} & 1/\sqrt{6}\\ 1/\sqrt{6} & -\sqrt{5/6}\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}x'\\ y'\end{bmatrix}$$

$$V=\frac{1}{2}\left(\frac{\sqrt{5}x+y}{\sqrt{6}}\right)^2+\frac{7}{2}\left(\frac{x-\sqrt{5}y}{\sqrt{6}}\right)^2=\frac{1}{2}(x')^2+\frac{7}{2}(y')^2$$

$$F_{x'}=-x' \qquad F_{y'}=-7y'$$

Ex 6.2.3 Degenerate

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$$H = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow \text{secular eqn} \quad \begin{vmatrix} -\lambda & 0 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda = 1, 1, -1$$

$$\lambda_1 = -1, \quad \vec{c}_1 = c \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\lambda_2 = 1, \quad \vec{c}_2 = c' \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \vec{c}_3 = c'' \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

6.3 Hermitian EVP

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$$H|c_i\rangle = \lambda_i |c_i\rangle \qquad H|c_j\rangle = \lambda_j |c_j\rangle$$

$$\langle c_j|H|c_i\rangle = \lambda_i \langle c_j|c_i\rangle = \langle Hc_j|c_i\rangle = \lambda_j^* \langle c_j|c_i\rangle$$

① if $i=j \Rightarrow \lambda_i = \lambda_i^*$

② if $i \neq j \Rightarrow \langle c_i|c_j\rangle = 0$

6.4 Hermitian matrix Diagonalization

$$Hc = \lambda c \Rightarrow HU^{-1}Uc = \lambda c \Rightarrow UHU^{-1}(Uc) = \lambda(Uc)$$

1) λ is invariant
2) $\exists U$: UHU^{-1} is diagonal $\Rightarrow \{\hat{e}_i\}$
3) ith eigenvector is ith column of U^{-1}

$$U^{-1}\hat{e}_i = \begin{bmatrix} | & & | \\ u_1^{-1} & \cdots & u_i^{-1} \\ | & & | \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \end{bmatrix} = \begin{bmatrix} | \\ u_i^{-1} \\ | \end{bmatrix}$$

Ex 6.4.1

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$$H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow \lambda = 2, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}; \lambda = 1, \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}; \lambda = -1, \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$U^{-1} = \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow \quad UHU^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Simultaneous Diagonalization

A, B commutes $\Leftrightarrow$ share eigenvector

Proof $\Leftarrow$ c_i eigenvector of A, B with eigenvalue a_i, b_i

$$BA\vec{c}_i = a_i B\vec{c}_i = a_i b_i \vec{c}_i$$
$$AB\vec{c}_i = b_i A\vec{c}_i = a_i b_i \vec{c}_i$$
$$\Rightarrow [A, B] = 0$$

Proof $\Rightarrow$ A, B commute, $A\vec{c}_i = a_i\vec{c}_i$

$A(B\vec{c}_i) = BA\vec{c}_i = a_i(B\vec{c}_i)$ $\Rightarrow$ $B\vec{c}_i$ is eigenvector of A $\Rightarrow B\vec{c}_i = b_i\vec{c}_i$

Spectral Decomposition

Wednesday, February 25, 2026 4:12 PM

H Hermitian

Spectral decomposition: $H = \sum_\mu |c_\mu\rangle \lambda_\mu \langle c_\mu|$

$$H^2 = \sum_{\nu\mu} \left(|c_\mu\rangle \lambda_\mu \langle c_\mu|\right)\left(|c_\nu\rangle \lambda_\nu \langle c_\nu|\right)$$

$$= \sum_\mu |c_\mu\rangle \lambda_\mu^2 \langle c_\mu|$$

$$\Rightarrow f(H) = \sum_\mu |c_\mu\rangle f(\lambda_\mu)\langle c_\mu|$$

$$H^{-1}|c_\mu\rangle = \frac{1}{\lambda_\mu}|c_\mu\rangle$$

$$\det\left(\exp(A)\right) = \prod_\mu e^{\lambda_\mu} = e^{\sum \lambda_\mu} = e^{\text{trace}(A)}$$

Expectation Values

Thursday, February 26, 2026 8:00 AM

$$\langle H \rangle = \langle \psi | H | \psi \rangle = a^\dagger H a = \sum_{\nu\mu} a_\nu^* H_{\nu\mu} a_\mu$$

$$= \sum_\mu |a_\mu|^2 \lambda_\mu$$

$$\langle a | a \rangle = \sum_\mu a_\mu^* a_\mu = \sum_\mu |a_\mu|^2$$

Positive Definite and Singular Operators

Positive definite: all $\lambda_i > 0$

Negative definite: all $\lambda_i < 0$

Singular : some $\lambda_i = 0$